

ABC MANUFACTURING CORPORATION

Environmental Health & Safety Department

STANDARD OPERATING PROCEDURE

Chemical Transfer Procedure

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Prepared By	J. Alvarez, EHS Specialist	Date	18 Dec 2025
Reviewed By	R. Nguyen, Sr. EHS Engineer	Date	07 Jan 2026
Approved By	M. Okafor, Director, EHS	Date	15 Jan 2026

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A. Document Control

A.1 Revision History

Rev	Date	Description of Change	Author	Approver
00	12 Sep 2022	Initial issue of the chemical transfer procedure, including bonding and grounding requirements.	J. Alvarez	M. Okafor
01	15 Jan 2026	Biennial review. Added risk matrix, transfer authorization permit, grounding verification log, and fill-limit controls.	J. Alvarez	M. Okafor

A.2 Distribution List

Copy No.	Holder / Role	Location	Format
Master	Document Owner — Manager, EHS	DMS (electronic master)	Electronic
01	Production Floor Supervisors	Line 1–4 supervisor stations	Controlled print
02	Chemical Store / Dispensing Lead	Chemical dispensing area	Controlled print
03	Maintenance Planner	Maintenance office	Electronic
04	Training Coordinator	Learning Management System	Electronic
05	Emergency Response Team Lead	ERT muster point	Controlled print

A.3 Controlled Copy Statement

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1. Purpose

This Standard Operating Procedure establishes the requirements for the safe transfer and dispensing of hazardous chemicals between containers at ABC Manufacturing Corporation. Its purpose is to control the hazards of chemical transfer — static-electric ignition of flammable vapors, spills and splashes, exposure, incompatible mixing, and overfilling — and to ensure transfers are authorized, bonded and grounded where required, and performed by trained personnel.

- Prevent static-generated ignition during the transfer of flammable and combustible liquids through correct bonding and grounding.
- Prevent spills, splashes, exposure, and incompatible mixing during dispensing and container filling.
- Ensure transfers are authorized, use approved equipment, and observe safe fill limits and rates.
- Support the site ISO 45001 OH&S and ISO 14001 environmental management systems.

2. Scope

This procedure applies to all ABC Manufacturing employees and contractors who transfer or dispense hazardous chemicals between portable and intermediate containers at any ABC Manufacturing facility.

2.1 Inclusions

- Manual and pumped transfers between drums, safety cans, portable containers, and intermediate bulk containers (IBCs/totes), including flammable-liquid transfer.
- Bonding and grounding, ventilation and ignition control, fill limits, and container labeling at the point of transfer.

2.2 Exclusions

- Fixed piped chemical distribution and bulk storage tank systems (Process Safety program) and bulk tanker loading/unloading.
- Compressed gas cylinder transfer (SOP-EHS-005) and line-breaking on energized systems, which additionally require Lockout/Tagout (SOP-EHS-011).
- Spill cleanup, addressed by SOP-EHS-003.

3. References

This procedure is based on publicly available regulations and consensus standards. Where a conflict exists, the most stringent applicable requirement governs.

Reference	Title / Subject
29 CFR 1910.106	Flammable liquids — including transfer, bonding, and grounding
29 CFR 1910.147	The control of hazardous energy (Lockout/Tagout) — line breaking
29 CFR 1910.1200	Hazard Communication — SDS and container labeling
29 CFR 1910.132–.138	Personal protective equipment
NFPA 30	Flammable and Combustible Liquids Code
NFPA 77	Recommended Practice on Static Electricity
NFPA 704	Identification of the Hazards of Materials for Emergency Response
ANSI/ISEA Z358.1	Emergency Eyewash and Shower Equipment
ISO 45001:2018 / ISO 14001:2015	OH&S and environmental management systems
Internal	SOP-EHS-001 Handling & Storage; SOP-EHS-003 Spill Response; SOP-EHS-005 Compressed Gas; SOP-EHS-011 LOTO

4. Definitions

Term	Definition
Bonding	Connecting two conductive objects (e.g., source and receiving containers) with a conductor so they are at the same electrical potential, preventing a static spark between them.
Grounding	Connecting a conductive object to earth so static charge is dissipated to ground.
Static Accumulation	Build-up of electrostatic charge generated by liquid flow, splashing, or mixing, which can discharge as an ignition-capable spark.
Flammable Liquid	A liquid with a flash point below 100 °F (Class I under NFPA 30 categories); readily forms ignitable vapor at ambient temperature.
Closed Transfer	A transfer method that minimizes vapor release and splashing, such as a sealed drum pump or dedicated coupling.
Safety Can	An approved container fitted with a spring-closing lid and flame arrestor for flammable liquids.
Fill Limit (Outage)	The maximum safe fill level that leaves headspace for thermal expansion of the liquid.
Line Breaking	Opening a pipe, line, or equipment that has contained a hazardous chemical; requires LOTO (SOP-EHS-011).

5. Responsibilities

Role	Responsibility
Area Supervisor	Authorizes transfers, verifies training and equipment, confirms controls are in place before flammable transfers.
Transfer Operator	Performs the transfer per this SOP, inspects and connects bonding/grounding, wears PPE, and stops on any abnormal condition.
Chemical Store / Dispensing Lead	Maintains approved transfer equipment, containers, and secondary containment; verifies product compatibility.
Maintenance	Maintains grounding points, ventilation, bonding/grounding cables, and the ground verification tester.
EHS	Approves transfer methods, audits compliance, supports incident investigation.

6. Required Training

Training Module	Audience	Frequency
Chemical Transfer Procedure (this SOP)	All transfer operators	Initial + on revision
Bonding & grounding of flammable transfers	Flammable transfer operators	Initial + annual
Approved transfer equipment & pump operation	Transfer operators	Initial + 2-yr refresh
HazCom / SDS & compatibility	All chemical handlers	Initial + annual
Spill prevention & response (SOP-EHS-003)	Transfer operators	Initial + 2-yr refresh

7. Required Qualifications

- Completion of the training in Section 6, verified in the LMS, and authorization by the Area Supervisor.
- Demonstrated competency in bonding/grounding connection and ground verification during on-the-job evaluation.
- Medical clearance to wear required PPE, including respiratory protection where the SDS or assessment requires it.

8. Required PPE

PPE is selected from SDS Section 8 for the product being transferred. The following is the minimum baseline for transfer tasks.

Transfer Task	Eye/Face	Hand	Body	Foot	Respirator
General transfer (sealed pump)	Safety glasses	Nitrile gloves	—	Safety shoes	—
Open pour / dispensing corrosives	Goggles + face shield	Chemical-resistant gauntlet	Chemical apron	Chemical boots	Per SDS
Flammable liquid transfer	Safety glasses / goggles	Nitrile gloves	FR-rated as required	Static-dissipative safety shoes	Per SDS

CAUTION

Wear static-dissipative footwear and avoid synthetic clothing that can generate static when transferring flammable liquids. Confirm glove breakthrough time against the SDS.

9. Required Equipment

- Bonding and grounding cables with clamps, and a ground verification tester (continuity/resistance) to confirm a sound ground before flammable transfers.
- Approved transfer equipment: drum pumps rated for the product, safety cans with flame arrestors, self-closing faucets/valves, and dedicated funnels.
- Secondary containment (spill pallet/tray) under the transfer point, sized to contain the larger container's contents.
- Local exhaust or general ventilation adequate to keep vapor below action levels; explosion-proof fixtures where flammables are transferred.
- ANSI/ISEA Z358.1 eyewash/drench shower within 10 seconds, Class B fire extinguisher, and a spill kit (SOP-EHS-003).

10. Hazard Identification

- Static-electric ignition** — charge generated by liquid flow/splashing igniting flammable vapor.
- Spill / splash** — release during pouring, pumping, or disconnecting, reaching personnel or drains.
- Chemical exposure** — skin/eye contact or inhalation of vapor during open transfer.
- Incompatible mixing** — transferring into a container holding a residual incompatible product.
- Overfill / expansion** — filling above the safe level, risking rupture or spill on thermal expansion.
- Mechanical / ergonomic** — pump handling, drum movement, and coupling/uncoupling.

11. Risk Assessment

Risk level is derived from the severity × likelihood matrix. Flammable transfers assessed as High or Critical require supervisor authorization and verified bonding/grounding before proceeding.

11.1 Risk Rating Matrix

Severity ↓ / Likelihood →	Rare	Unlikely	Possible	Likely	Almost Certain
Catastrophic	Medium	High	High	Critical	Critical
Major	Medium	Medium	High	High	Critical
Moderate	Low	Medium	Medium	High	High

Severity ↓ / Likelihood →	Rare	Unlikely	Possible	Likely	Almost Certain
Minor	Low	Low	Medium	Medium	High
Negligible	Low	Low	Low	Low	Medium

11.2 Task Risk Register

Hazard	Severity	Likelihood	Risk Level	Control Measures
Static ignition during flammable transfer	Catastrophic	Unlikely	High	Bond source-to-receiver, ground assembly, verify ground, low fill rate, submerged fill, no ignition sources.
Splash/spill during open pour	Moderate	Likely	High	Closed transfer/pump, funnel, secondary containment, controlled rate, goggles/face shield.
Vapor inhalation during transfer	Major	Possible	High	Ventilation/local exhaust, respiratory protection per SDS, minimize open time.
Transfer into incompatible residue	Major	Unlikely	Medium	Use dedicated/clean containers, verify contents, never reuse without triple-rinse/label.
Overfill / expansion rupture	Moderate	Possible	Medium	Observe fill limit/outage, do not fill unattended, use fill gauge/stop.

12. Detailed Step-by-Step Procedure

12.1 Pre-Job Preparation & Authorization

1. Obtain the SDS for the product; confirm hazard class, PPE (§8), incompatibilities (§7/§10), and any special transfer precautions.
2. Obtain transfer authorization from the Area Supervisor for flammable transfers or any transfer rated High/Critical (Appendix C).
3. Confirm the receiving container is correct, clean, compatible, and correctly labeled (or label immediately after filling per SOP-EHS-002).
4. Position the transfer over secondary containment; confirm ventilation, eyewash/shower, extinguisher, and spill kit are available and serviceable.

12.2 Equipment Inspection

1. Inspect pump, hoses, couplings, funnels, and safety-can flame arrestor for damage, leaks, and compatibility; reject defective equipment.
2. Inspect bonding/grounding cables and clamps for clean, tight, conductive contact points (bare metal, free of paint/corrosion).

12.3 Bonding & Grounding (Flammable / Combustible Transfers)

1. Bond the source container and the receiving container together with a bonding cable, clamping to clean bare metal on each.
2. Ground the bonded assembly to a verified building/earth ground point.
3. Verify the ground with the ground tester (resistance within the site limit) and record it on the grounding verification log (Appendix B).
4. Maintain bonding and grounding throughout the transfer. Remove the grounding connection LAST, after closures are replaced and flow has stopped.

WARNING

Never transfer a flammable liquid without verified bonding and grounding. A static spark from unbonded containers can ignite vapor and cause a flash fire or explosion.

12.4 Ventilation & Ignition Control

1. Confirm ventilation/local exhaust is operating; keep the transfer point clear of ignition sources (open flames, sparks, hot work, non-rated electrical).
2. Prohibit smoking and non-rated tools/devices in the transfer area; post the area during flammable transfers.

12.5 Work Execution — The Transfer

1. Open containers slowly to relieve pressure. Begin transfer at a low flow rate to limit static generation and splashing.
2. Use submerged/bottom fill or a dip tube where practical; keep the nozzle/spout in contact with the receiving container.
3. Observe the fill limit (outage) to leave headspace for thermal expansion; never fill above the safe level and never leave a transfer unattended.
4. Stop transfer immediately on any leak, spill, static discharge, unusual odor, or equipment abnormality and investigate before continuing.

12.6 Verification & Disconnect

1. On completion, stop flow, allow a brief relaxation time for static to dissipate, then close and reseal both containers.
2. Disconnect the transfer equipment; remove the bond, then remove the ground LAST. Return cables and tester to storage.
3. Confirm the receiving container is correctly labeled and the fill level is within limits.

12.7 Housekeeping

1. Wipe residue from equipment and surfaces; manage any incidental spill per SOP-EHS-003 and dispose of contaminated materials as hazardous waste.
2. Return chemicals to compatible storage; restock spill kit/PPE; leave the area clean with clear eyewash and extinguisher access.

12.8 Documentation

1. Complete the transfer authorization (Appendix C) and grounding verification log (Appendix B); file in the DMS per the records-retention schedule.

12.9 Escalation Procedures

1. Static discharge, vapor build-up, or equipment failure during flammable transfer → stop, disconnect safely if possible, evacuate, and notify the supervisor.
2. Spill beyond containment or any exposure → initiate Emergency Response (Section 13) and SOP-EHS-003.

13. Emergency Response

WARNING

On any flammable-transfer fire or vapor ignition, stop flow if safe, evacuate, activate the alarm, and summon the ERT. Do not attempt to fight a spreading fire.

13.1 Immediate Actions

- Stop the transfer and source if safe; eliminate ignition sources; evacuate personnel upwind and isolate the area.

13.2 Medical Response

- For exposure, flush at the eyewash/drench shower for at least 15 minutes and seek medical aid; provide the SDS to responders.

13.3 Fire Response

- Use a Class B extinguisher on an incipient fire only if trained; otherwise activate the alarm and evacuate.

13.4 Spill Response

- Contain and clean an incidental spill per SOP-EHS-003; escalate larger releases to the ERT.

13.5 Equipment Shutdown

- Isolate pump power and close valves feeding the transfer where safe; do not re-enter an unsafe area to shut down.

13.6 Notification Chain

Step	Notify	Contact / Method	When
1	Area Supervisor	Radio / floor phone	Immediately on any release, fire, or exposure
2	EHS On-Call	Ext. 2400 / site radio	Within 5 minutes
3	Emergency Response Team	Emergency line 2222	Fire, injury, or release beyond containment
4	External Emergency Services	911	Life threat, fire, or major release
5	EHS Manager & Plant Manager	Per call tree	Within 30 minutes; recordable/reportable events

14. Required Documentation

Record	Purpose	Owner	Retention
Transfer Authorization / Permit (App. C)	Authorize and record flammable/high-risk transfers	Area Supervisor	3 years
Grounding Verification Log (App. B)	Evidence of verified ground before flammable transfer	Transfer Operator	3 years
Equipment Inspection Checklist (App. D)	Pre-transfer equipment condition	Transfer Operator	3 years
Training Records (LMS)	Evidence of qualified operators	Training Coordinator	Duration + 3 years
Incident / Spill Report	Releases, exposures, ignitions	EHS	5 years

15. References (Public Standards)

- OSHA 29 CFR 1910.106 — Flammable liquids (transfer, bonding, grounding).
- OSHA 29 CFR 1910.147 — Lockout/Tagout (line breaking).
- OSHA 29 CFR 1910.1200 — Hazard Communication.
- NFPA 30 — Flammable and Combustible Liquids Code.
- NFPA 77 — Recommended Practice on Static Electricity.
- NFPA 704 — Identification of the Hazards of Materials for Emergency Response.
- ANSI/ISEA Z358.1 — Emergency Eyewash and Shower Equipment.
- ISO 45001:2018; ISO 14001:2015 — Management-system standards.

16. Appendices

Appendix A — Bonding & Grounding Connection Sequence

- STEP 1 — Confirm both containers and clamps present clean bare metal contact points (no paint, rust, or coating).
- STEP 2 — BOND: clamp a bonding cable from the source container to the receiving container.
- STEP 3 — GROUND: connect the bonded assembly to a verified building/earth ground point.
- STEP 4 — VERIFY: check ground resistance with the tester; record on the grounding log (Appendix B). Do not proceed if out of limit.
- STEP 5 — TRANSFER at low rate; keep bonding and grounding connected throughout.
- STEP 6 — DISCONNECT after flow stops and closures are replaced: remove the bond, then remove the GROUND LAST.

Appendix B — Grounding Verification Log

Date / Time	Product	Ground Reading	Within Limit?	Operator
			☐ Yes ☐ No	
			☐ Yes ☐ No	
			☐ Yes ☐ No	
			☐ Yes ☐ No	

Appendix C — Chemical Transfer Authorization / Permit

Field	Entry
Permit No. / Date	
Product / SDS reference	
From container → To container	
Flammable / combustible?	☐ No ☐ Yes (bonding & grounding required)
Bonding & grounding verified	☐ N/A ☐ Verified (log ref.)
Ventilation & ignition control confirmed	☐ Yes
PPE confirmed (per SDS §8)	☐ Yes
Secondary containment in place	☐ Yes
Operator name / signature	
Authorized by (Supervisor) / date	

Appendix D — Pre-Transfer Equipment Inspection Checklist

#	Item	OK	Action Needed
1	Pump/hoses/couplings intact, compatible, leak-free	☐	☐
2	Safety-can flame arrestor present and clean	☐	☐
3	Bonding/grounding cables & clamps clean and conductive	☐	☐
4	Ground tester functional / in calibration	☐	☐
5	Secondary containment in place and serviceable	☐	☐
6	Ventilation operating; ignition sources controlled	☐	☐
7	Eyewash/shower, extinguisher, spill kit accessible	☐	☐
8	Receiving container correct, clean, compatible, labeled	☐	☐